...wer and mark it in the answer sheet above

1. The space needed by a program is the space reserved for its variables in memory only.

A. Correct B. Incorrect

2. Consider the following struct; Sizeof(e) is:

```
struct Employee {
    int code;
    char name[5];
}e;
```

A. 9 bytes B. 12 bytes C. 16 bytes

3. Graphs are considered ________.

A. Linear Lists B. Non-Linear Lists

4. The hiding of the data structure implementation inside the ADT is referred to________ .

Abstraction B. Encapsulation

...tack is called ________.

B. LILO C. FILO D. None of the above

...ed lists are always using less memory than ...-based lists regardless of the data type of ...current list size, nor the list maximum size.

B. No

7. Which one of the following is an application of Stack Data Structure?

A. Toll Station
B. Check-out in Big supermarket
C. Printer
D. Evaluation of Arithmetic Expressions

8. If the sequence of operations - push (3), push (5), pop, push (3), push (5), pop, pop, pop, push (5), pop is performed on a stack, the sequence of popped out values would be ________.

A. 5,3,3,5,5 B. 5,5,3,5,5
C. 5,3,5,5,3 D. 5,5,3,3,5

EDCB

9. The five items: A, B, C, D, and E are pushed in a stack, one after another starting from A. The stack is popped four items and each element is inserted in a queue. Then two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is________.

ABED

A. B B. C D. D

D

10. Consider stack with array-based implementation, popped item from the sta... kept in the array.

A. Yes B. No

11. Consider the following sequence of operations on an empty stack.
push(3); push(4); pop(); push(5); push(6); s=pop();
Consider the following sequence of operations on an empty queue.
enqueue(7); enqueue(8); dequeue(); enqueue(9); enqueue(10); q=dequeue();
The value of s + q is ________.

A. 16
B. 14
C. 11
D. 8

12. Which of the following is a valid function prototype for a stack pop operation?

A. void Pop(EntryType item, StackType s);
B. EntryType Pop(StackType s);
C. Both A & B
D. None of the above

13. What does the following function do?

```
void fun(Queue *Q){
    Stack S;        entryType    elm;
    create(&S);
    while (!isEmpty(Q)) {
        deQueue(Q, &elm);
        push(&S, elm);  }
    while (!isEmpty(&S)) {
        pop(&S, &elm)
        enQueue(Q, elm);  } }
```

A. Removes the last from Q
B. Makes Q empty
C. Reverses the Q
D. Keeps the Q same as it was before the

14. Implementing QUEUES on a circular array is more efficient than implementing QUEUES on a linear array with shifting all items to front when dequeue operation.

A. Yes
B. No

15. In queues based on circular array implementation, if front==rear this means ____.

A. The queue is empty
B. The queue is full
C. The queue has exactly one element
D. None of the above

16. When inserting a new element in the following list, the allowed insertion range is ________.

A. 0 <= pos <= n
B. 0 <= pos <= n+1
C. 0 <= pos <= Max-1
D. 0 <= pos <= Max

17. What does the following function do for a given Linked List with first node as head?

```
void fun1(struct node* head)
{
  if(head == NULL)
    return;

  fun1(head->next);
  printf("%d  ", head->data);
}
```

A. Prints all nodes of linked lists
B. Prints all nodes of linked list in reverse order
C. Prints alternate nodes of Linked List
D. Prints alternate nodes in reverse order Linked

18. When implementing Linked Stack and Linked Queue as layers above Linked List ADT, which function of the following has the least performance.

A. push
B. pop
C. enqueue
D. dequeue

19. In general, inserting / deleting an element to/from a list is faster in ________.

A. Array-based List Implementation
B. Linked List Implementation

20. Consider an implementation of unsorted array-based list. Which of the following operation is the simplest?

A. Insertion at the front of the list
B. Insertion at the end of the list
C. Both are the same

With my best wishes
Dr. Mohammed El-S